

Let's Talk Tempo: Variable Speech Rate Conversational Agents

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Abstract

Despite strides in making Conversational Agents (CAs) more human-like through pitch variation, many users still find such agents difficult to communicate with due to assumptions made by developers on what "normal" speech patterns are. Rate of speech is one such variable that remains inflexible in most CAs despite its role in governing the comprehension and efficiency of interpersonal communication. We aim to implement and test a dynamic conversational agent that adapts to the user's rate of speech. The findings will reveal whether changing the speech rate of a CAs to match that of the speaker impacts the user experience, evaluated for both comprehension and efficacy.

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1 Goal

Currently, Commercial Conversational Agents are often calibrated singularly to an "average" user in terms of their rate of speech. We seek to test our hypothesis that the use of an adaptive Speech Rate (SR) CA that adjusts based on the user's SR will improve comprehension and efficiency.

2 Related Work

For English, four main paralinguistic aspects or adjustable elements of speech are identified: speech rate, volume, pitch, and timbre[3]. In the context of interacting with CAs, these factors can be modulated from interaction to interaction as best suited to the needs of the speaker without inherently modifying the content. Variations in speaking rate (SR) preferences can be seen across different speaker populations. A Native speaker (NS) and non-native speaker (NNS) of a language, for example, are impacted differently by the rate of speech when it comes to speech and listening comprehension. Preliminary experiments by Inoue and Liao to assess the impact of speech rate (SR) on comprehension revealed that of the 115 incomprehensible speech segments recorded in 150 minutes of conversation, 24 cases were due to the SR of the NS being too fast, comprising 21% of all cases [4]. When examining how such factors impact interaction with speech synthesizing systems, researchers

evaluating the use of CAs by brands found that consumers who used digital assistance with a fast or slow SR were less likely to want to use voice-assistive technology than consumers who interacted with a voice agent who used a "moderate" SR [3].

Another study showed that manipulating the speech rate could impact the user's impressions of the agent. Users felt that the base rate agent was more human-like and intimate, especially evident in their engagement in small talk with the agent. In contrast, users described the faster agent as mechanical and found it frustrating when they would miss information. This changed their usage such that they used more functional commands. [2].

Another factor to consider when increasing the accessibility and efficacy of dialogue is that people are often able to comprehend synthetic speech at a faster rate than they can speak it themselves. This is especially true for those who have been exposed to synthetic speech from an earlier age who may find it more efficient to communicate with a CA that speaks faster than themselves [1]

3 Contribution

The existing studies on voice agents described above have been limited to manipulating the agent's rate of speech at predetermined scales. Choi et. al exclusively tested an increased SR against a baseline, with findings that were primarily applied to compare the preferences of blind/low-vision users of the voice agent and fully-sighted users.

Rather than providing unidirectional or static adjustments to speaking rate (i.e. 1.75x speed) in the way prior research has, our aim is to implement a proof-of-concept for an adaptive VA, in a small user study for feedback and impressions. By comparing the feedback and data to the default SR of the system, this paper will help determine if the adaptive SR of the agent is positively or negatively impacting the user experience and listening comprehension, or if there is no notable difference.

4 Implementation

As outlined in Figure 1, it is first a well-established means of speech-to-text transcription as well as speech rate detection were identified. This manifested in using industry standard speech-to-text systems, and PRAAT scripts from the Myprosody package and repository. Our python notebook acts as a wrapper for our chosen LLM. We will also discuss various challenges that were encountered in this section for the two approaches taken, followed by details of shared components between them.

4.1 Local approach

While a local approach ended up as the proof-of concept, setting up our environment in figuring out the right models and compute resources that were available was troublesome. There were many times where the Linux computer used would run out of memory or

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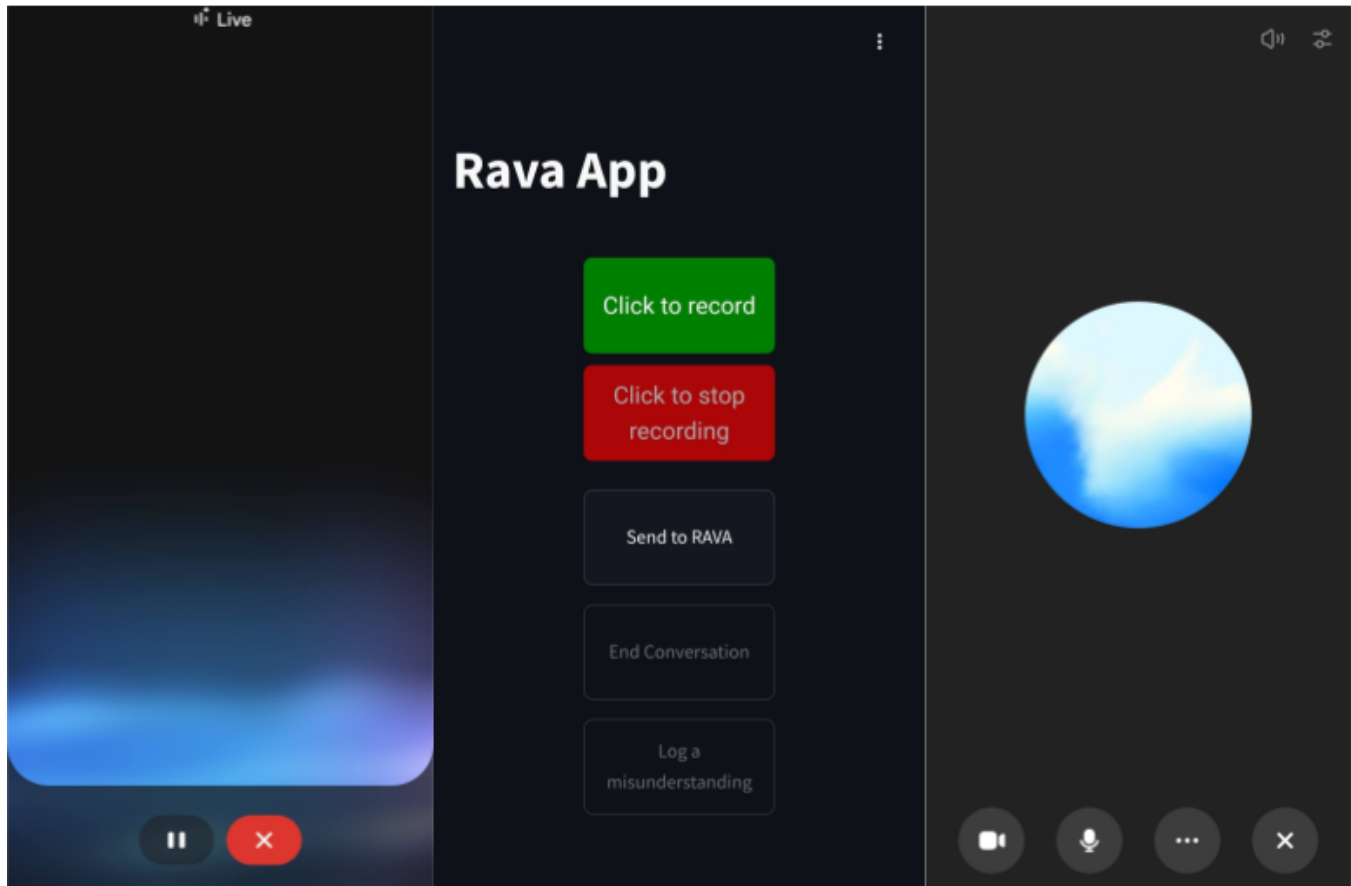


Figure 1: From left to right: Gemini Live, Our web-app approach, RAVA <https://github.com/i-am-ami/hci-rava-project/tree/deployable>, and ChatGPT Advanced Voice

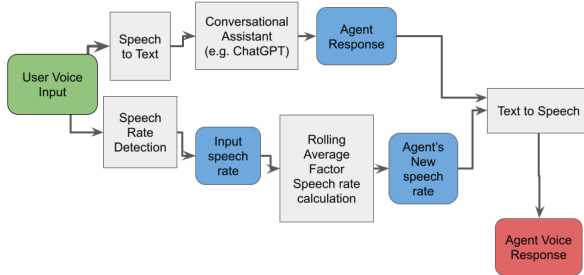


Figure 2: Low-Fidelity Sketch pipeline flow. Green is for User input, Blue for Intermediate Data, Red for Output, Square Boxes denote Systems/Programs, and the Rounded Boxes denote Data.

encounter driver errors that made the system unstable. The only way was to manually address these issues as they came up, especially for CUDA programs. Ollama was used to source LLM models,

specifically mistral:7b, qwen2.5:0.5b, and llama3.2:1b. Testing of the models sped up drastically with the latter 2 models, as they were both small enough to fit on our GPU's limited VRAM (4GB). Llama3.2:1b was the model used in the final code, as Qwen had the habit of occasionally hallucinating responses in Chinese instead of English, on top of the overall poorer quality of responses.

We initially tried using Google's gTTS python package to speak the LLMs responses, but this involved manually speeding up playback of the agent's response. This produced an unnatural tone of voice when any speedup/slowdown of speech were applied. We transitioned to Piper-TTS, which had built-in functionality to change its playback speed, without any degradation in playback quality.

4.2 Deployable Web-App approach

As the project stretched beyond the fall semester, development pivoted to an Azure cloud-based approach. By making use of Azure's Speech Recognizer, integration of ChatGPT models via Azure AI Studio, its own TTS system with playback speed manipulation, this approach was pursued in the hope of being able to deploy the app and conduct user testing remotely. Sadly, this never happened. Difficulties and limitations of using a Python Web App library, Streamlit, for rapid implementation created many difficulties in

creating the functional app in tandem with Python's poor ability to handle multiple interactions asynchronously. One by one these challenges were overcome, but at the cost of time. As time ran out, we pivoted back to the local setup, which we had been steadily back-porting features to in preparation for such a scenario.

4.3 Speech Rate

Common between both approaches is the usage of Myprosody . It works best with 48kHz .wav files, to ensure that the syllables per second can be extracted from the file. Challenges arose from integrating this package, as its code only outputs values to the terminal. There were also ambivalent error messages on the audio quality that are printed if the program encounters any error in the process of processing the file, which includes file path mismatches, such as trying to use relative paths.

After all of that was sorted, we came up with the following calculation to make the system vary its speech rate to the user:

$$\alpha = \frac{U_{\text{current}}}{U_{\text{default}}}, \quad S_{\text{new}} = \frac{\alpha + S_{\text{current}}}{2}, \quad \beta = \frac{T}{U_{\text{default}}} * W$$

$$\text{length_scale (needed by PiperTTS)} = \frac{\beta}{S_{\text{new}}}$$

The U variables denote the user speech rates in syllables per second, while S variables denote the speed factor. based on an average of the current speed factor and a new factor based on how the user's current speech rate U_{current} is in comparison to the default, U_{default} . T is the default speech rate of the PiperTTS. We have to apply the speed factor's inverse to the ratio between the User's and PiperTTS's default speech rate due to how PiperTTS implements speed via a length_scale variable, where smaller rationals make the response faster, and larger ones make it slower. The aforementioned T/U_{default} fraction is used to account for the difference between the User's default and the TTS's, and W is a little bit of manual tuning we did to speed up the TTS to match the user's default syllables per second of 3.0, when the speed factor is just 1.0.

A while-true loop keeps the conversation going, and usage of Langchain packages facilitate the prompt for the agent as well as message history. This pipeline does not have online search capabilities.

Both approaches also have built-in logging functionality, and export .csv logs with a ISO format timestamp, message, and current speed factor as the columns. Messages are to log what the user's input speech rate is, what the new speed factors are, and responses for the user and agent.

5 User Testing Setup

Since this work leaned further to a system rather than a user study as was initially proposed, the setup and data collected on participants was slightly scaled down to suite the situation more appropriately.

Collected quantitative data consists from the users was the .csv log files of their session with the agent in free unstructured conversation for about 3-4 minutes. These log files will be used to determine the user's min, max and average speech rate, as well as

the same for the assistant, as well as percentages on how much faster/slower the adaptive rate response is to the static rate.

Apart from that, qualitative user responses were noted from an informal debrief on their experience after they had finished using the agent, after a 5-minute break. Questions in the interview included open-ended feedback, as well as their proficiency and history with CAs and voice-assistive technologies (STT and TTS). Further follow-up questions ensued as to explore their experience further, the note-worthy responses which will be stated and discussed in the Results section.

As for the operations during testing, all participants were present in a waiting room, where a separate vacant room was set up to for them to use the agent in, making use of the laptop and the notebook. In simpler terms, participants waited in the living room of one of our apartments, and a vacant bedroom was used for the testing room.

Users could see the transcription of their speech, but not see the text of the Agent's response. One of use was present, as an observer in the testing room and to handle the participants. Users made use of a headset with a boom microphone to ensure clear audio.

Participants were informed to speak at a natural rate of speech, and that both systems were commercially available in an attempt to avoid biases towards the implemented system since it was created by one of the observers. Users were not explicitly told that the agent was designed to adapt to their speech rate, and were only told after testing to illicit more informative feedback/responses.

for the informal debrief, all participants were together in a discussion, allowing each other to help refine their feedback and comment on each other's experiences.

6 Results

6.1 Quantitative Results & Discussion

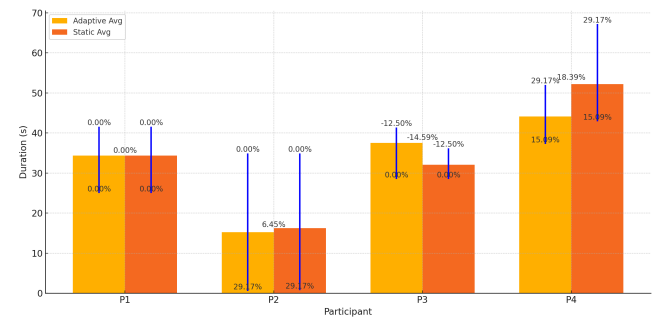


Figure 3: Analysis of the Agent Speech Duration. The bars indicate the difference in Average duration for the Adaptive Speech rate (left bar, yellow) and the Static Speech Rate of 3 syllables per second. The blue lines on each bar indicate the min and max percentage difference between the Static and the Adaptive min and max durations, where negatives indicate the Adaptive was slower, and positive percentages indicate that it was faster. The percentage in the middle of the two bars is the percentage difference of the average duration, following the same positive/negative scheme as with the min/max values.

To determine "quantitatively" whether or not our system met the criteria of improving the efficiency and comprehension of conversation, we will take a look at the metrics that are illustrated in Figure 3. In the case of P2 and P4, the Adaptive SR responses, on average were faster, albeit at small margins. This is inline with would be expected with the calculations the code makes to adapt SR, always erring on the safe side and being slightly slower than the user's speech rate when they start speaking faster. P3 had the exact opposite relationship between the average agent response duration and the percent difference, as the adaptive SR did not slow down as much as the user, still erring on the safe side as well.

At least from a quantitative perspective, there is a small improvement in communication efficiency here. For the users like P4 who can speak faster, having a CA that can keep up with you is certainly going to improve the efficiency of the interaction, at east in terms of time. In terms of observations, P4 was generally more energetic than the other participants. That being said, it should be noted that for the average users, like P2 and P1, there will most likely not be any significant improvements in efficiency.

6.2 Qualitative Feedback & Discussion

The feedback and interview responses from the users tell a similar story in terms of magnitude of impact, but in the opposite direction. For instance, P4 noted that hearing the CA speed up, even in its more gradual manner, made the conversation feel more tense, as they felt the need to keep up the pace, and that at a certain point they most likely would have had to slow down further in a "crashing" manner, essentially walking back some of the time savings that they would have gained had the conversation continued. P2 stated they also noticed the speed up of the assistance, but didn't state it aloud in their feedback this until we had. P1, as would be expected for someone who spoke at a consistent rate, didn't have much in terms of input on the adaptability of the CA.

Comprehension however was noted as generally poorer than prior interactions with other conversational agents, as stated by P2 and P4. when we inquired more about these experiences, P4 mentioned how they would often use PDF readers to help get through research papers, allowing them to read along with voice. P2 had mentioned that they often watch YouTube videos at higher speeds using an extension to manipulate the video speed in a granular manner. they also had brought up the usage of captions at times as well to aid their comprehension, especially in busier environments. P3 had noted that they had noticed the assistant slowing down, but didn't care much for the difference. This does line up with the results in the work of Choi et al, where slowing down the assistant speaking rate, even though it created a more comfortable experience didn't exactly improve comprehension, and worsened the efficiency of the conversation as well [2]. Users can typically read faster than they can hear.

The .csv tables used to create Figure 2 will be in the lets-talk-tempo GitHub repository: <https://github.com/AbhinavJayaraman/lets-talk-tempo/tree/final>

7 Future Work & Considerations

There were many aspects of this system that were compromised on for the sake of time, such as participant size, deploying the web app,

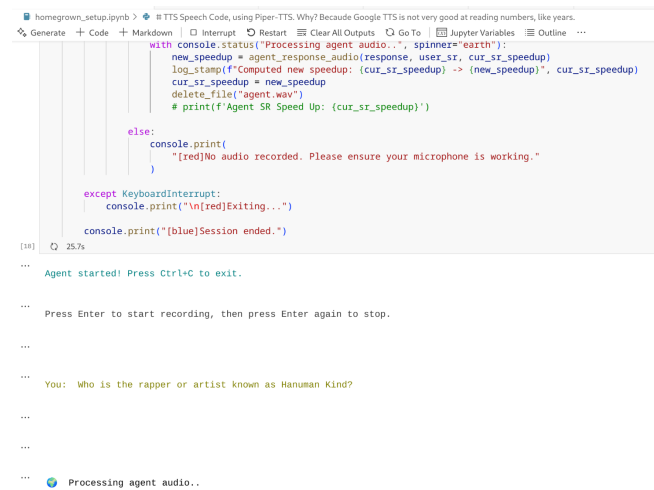
or experimenting with other adaptive speech rate equations. The former is where we would like to expand on this work, to see what our small sample size captured and missed. Possibly comparing the inclusion of the agents responses could be fruitful to test to verify that impact further.

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8 Appendix

8.1 Picture of the Python Notebook with an example user query



```

homegrown_setup.ipynb | TTS Speech Code, using Piper-TTS. Why? Because Google TTS is not very good at reading numbers, like years.
Generate | Code | Markdown | Interrupt | Restart | Clear All Outputs | Go To | Jupyter Variables | Outline | ...

with console.status("Processing agent audio...", spinner="earth"):
    new_speedup = agent_response_audio(response, user_sr, cur_sr_speedup)
    log_stamp(f"Computed new speedup: {cur_sr_speedup} -> {new_speedup}")
    cur_sr_speedup = new_speedup
    delete_file("agent.wav")
    # print(f'Agent SR Speed Up: {cur_sr_speedup}')

else:
    console.print(
        [red]No audio recorded. Please ensure your microphone is working.
    )

except KeyboardInterrupt:
    console.print("\n[red]Exiting...")
    console.print("[blue]Session ended.")

[18] Q 25.5%

... Agent started! Press Ctrl+C to exit.

... Press Enter to start recording, then press Enter again to stop.

...

... You: Who is the rapper or artist known as Hanuman Kind?

...

... Processing agent audio..

```

8.2 Local Setup Computer Specifications

The system we made use of was an Acer Swift x 14-in Laptop (2021), running Fedora Linux Workstation 41. for better Wi-Fi stability, the default MT7921 WiFi card was replaced with a better supported but older Intel 7265 card. Power-saving measures were turned off for the card. The laptop was running off a 256 GB SSD, with 16 GB of RAM. The CPU is the AMD RYZEN 7 5800u with 8 cores and 16 threads, and a max power draw of 28 Watts. The GPU was the Nvidia RTX 3050 TI, with 4 GB of VRAM, with a max power draw of 40 Watts.